Emmanouela Tsakalidou

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Software Engineer - DevOps, CI/CD, Backend & Systems

Profile

Software Engineer focused on DevOps, CI/CD automation and reliable software delivery, with hands-on project experience across Jenkins, Docker, Linux, Java/Spring Boot, Python, TypeScript, backend APIs, automated testing and troubleshooting. Interested in roles combining software development with CI/CD infrastructure, automation and engineering reliability.

Technical Skills

DevOps & CI/CD : Jenkins, CI/CD, Docker, Linux/UNIX, Bash, Shell Scripting, Git, GitHub
Backend & Development : Java, Spring Boot, Python, TypeScript, JavaScript, Node.js, Express, FastAPI, REST APIs, C, C++, C#, Kotlin
Testing & Quality : JUnit, pytest, Jest, Playwright, Postman, Ruff, API Testing, Smoke Testing, Debugging, Defect Investigation
Databases : PostgreSQL, MySQL, SQLite, MongoDB, SQL, Prisma, Mongoose, SQLAlchemy

Experience

Software Developer Intern February 2025 - September 2025
Valmore *Athens, Greece*

- Developed, tested and maintained application features within agile development cycles using React Native
- Wrote and executed Jest tests to improve test coverage and identify software defects earlier in the development process
- Reproduced and investigated application defects using structured troubleshooting and root-cause analysis
- Built reusable React Native components and validated behaviour, consistency and feature integration

Projects

PipelineForge | *Jenkins, Docker, Java 21, Spring Boot, Gradle, JUnit, Linux, Bash* 2026

- Built a Jenkins Pipeline as Code workflow automating source checkout, environment verification, Gradle compilation, JUnit testing and Spring Boot JAR packaging
- Configured distributed Linux build agents and a dedicated Docker build agent to separate application builds from container-image creation
- Built versioned Docker images using Jenkins build metadata and Git commit information, followed by automated runtime container smoke tests
- Introduced controlled CI and runtime failures to verify quality gates, investigate root causes and confirm recovery through repeated pipeline runs

Jenkins CI Reliability Lab | *Jenkins, Docker, Linux, Python, pytest, Ruff, Shell* 2026

- Created a local distributed Jenkins environment with a persistent Dockerized controller and two labeled agents supporting Pipeline from SCM and parallel CI execution
- Configured pipeline timeouts, targeted retries, failure classification, build retention, artifact archiving and queue/executor capacity experiments
- Simulated infrastructure and agent failures to practise CI troubleshooting and distinguish product, dependency and infrastructure failures
- Developed Python and shell monitoring tooling using the Jenkins REST API to inspect agent status, queue activity, build results, execution duration and system health

PriceWise - BSc Thesis | *TypeScript, React, Node.js, Express, PostgreSQL, Prisma, Docker* 2026

- Developed a full-stack grocery price-comparison platform with REST APIs, PostgreSQL data models, multilingual React interfaces and real-time updates
- Built scheduled data-collection workflows to retrieve, validate, normalize, deduplicate and persist supermarket product data from multiple external sources

- Added WebSocket updates, database constraints, automated scraper workflows and source-level error handling to improve data reliability

C++ Sensor Dashboard | *C++17, CMake, Multithreading, HTTP, JSON* 2026

- Developed a multithreaded C++17 sensor-monitoring application with simulated sensor data, HTTP endpoints and live browser-based visualization
- Used `std::thread`, `std::mutex`, `std::lock_guard` and `std::atomic` to coordinate thread-safe shared state and concurrent processing
- Applied RAII, smart pointers, STL containers, configuration validation and structured error handling

RepoSentry | *Python, Typer, HTTPX, pytest, mypy, Ruff* 2026

- Developing a Python CLI for analysing GitHub repository health, including repository metadata, issues and pull-request information
- Added automated pytest coverage, strict mypy type checking and Ruff linting/formatting as development quality gates

Education

University of Essex September 2024 - June 2026
BSc (Hons) Computing (Software Development) *Overall Grade: 73%*
IVT OMIROS September 2023 - June 2024
Web Designer/Developer - Video Games

Selected Coursework

Operating Systems and Network Programming

- Studied Linux/UNIX, C, processes, threads, CPU scheduling, synchronization, virtual memory, file systems and TCP networking
- Built a concurrent TCP file-transfer application in C using `fork()` and client-server sockets

Advanced Programming

- Developed a REST API using Node.js, Express and MongoDB with JWT authentication, middleware, filtering, sorting, pagination, aggregation and structured error handling

Object-Oriented Programming

- Applied C#/.NET and object-oriented programming concepts alongside UML, recursion, data structures, algorithms and Big-O analysis

Certifications

React Native - The Practical Guide - Udemy, 2025
Learn SQL Course - Codecademy, 2024
Responsive Web Design - freeCodeCamp, 2023